Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination-June 2022
Course Code: EEE 421, Session: 2018-19
Course Title: Solid State Devices

Full Marks: 100 **Credits: 3.0** **Exam Durations: 3 Hours**

Part – A

*[(Answer any **TWO** sets of questions)]*

1. (a) Explain the temperature dependance [*sic*: should be "dependence"] of carrier concentration of an extrinsic semiconductor with the help of graph. 7
   (b) A germanium sample is doped with 10^{15} boron atoms per cm^3. Find the electron concentration. Intrinsic carrier concentration of germanium is $2.5 \times 10^{13}/cm^3$ at $300k$. 6
   (c) At absolute zero temperature, what is the maximum energy of an electron in a semiconductor material? 5
   (d) What is the Hall effect? Derive the expression for finding the carrier concentration of a semiconductor from Hall voltage. 7

2. (a) What is storage delay time? What is the impact of reverse recovery time of a junction in switching application? 7
   (b) What is Fermi energy? kT is equal to how many eV at $300k$? 6
   (c) What is Einstein Relation? Derive the expression. 6
   (d) Wat [*sic*: should be "What"] is the depletion capacitance of a PN junction? Explain its variation with reverse bias voltage. 6

3. (a) With brief description sketch the energy band diagram of a MOS structure at different bias voltage. (Use no bias condition, positive bias voltage and negative bias voltage). 8
   (b) What are the parameters that affect the magnitude of threshold voltage in a MOSFET? 6
   (c) What is body effect of MOSFET? 5
   (d) Explain the amplification action of a transistor. 6

Part – B

*[(Answer any **TWO** sets of questions)]*

4. (a) Sketch the ideal electron and hole current components through a pn junction under forward bias. 7
   (b) What are the factors which cause base current in a transistor? 3
   (c) Differentiate between ohmic and rectifying contacts. 4
   (d) What is base width modulation? 5
   (e) Explain the working of Tunnel diode. Draw its characteristics curve. 6

5. (a) Describe diffusion process. Derive the expression for diffusion current density. 7
   (b) Draw the ideal low-frequency capacitance versus gate voltage curve of a MOS capacitor with p-type substrate and indicate different regions. 5
   (c) Show that diffusion length is the average length a carrier diffuses before recombination. 6

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(d) A silicon n-channel JFET at $T = 300K$ has a gate doping concentration of $N_a = 10^{18} cm^{-3}$ and a channel doping concentration of $N_d = 2 \times 10^{16} cm^{-3}$. Determine the metallurgical channel thickness, a, such that the pinchoff voltage is $V_p = -2.50\ V$. 7

6. (a) The parameters of an n-channel silicon JFET are $\mu_n = 1000\ cm^2/V-s$, $N_d = 5 \times 10^{15} cm^{-3}$, $a = 0.50\ \mu m$, and $L = 2\ \mu m$. Determine the cutoff frequency. 6

(b) Design the ratio of emitter doping to base doping in order to achieve an emitter injection efficiency factor of $\gamma = 0.9967$. Assume a *npn* bipolar transistor, and also consider $D_E = D_B$, $L_E = L_B$ and $x_E = X_B$. 6

(c) What is current crowding? What is the impact of current crowding in high power devices? 7

(d) With suitable energy band diagrams, explain Schottky contact. 6

Class Test 01 (its questions 1 to 4 are 7(a) to 7(d))

[Class test-01. Solid State Device. Time-30min. Marks: 10]

7. (a) Define the density of states and carrier concentration. 2

(b) Explain with the help of equations and graphs- why intrinsic carrier concentration (n_i) for a typical semiconductor is high (~10^6-10^9/cc) at 300K even though the Fermi Dirac distribution is low at that temperature. 3

(c) What is electron-effective mass? Two possible conduction bands in the E versus k diagram are given in Figure 1. State which band will result in the heavier electron effective mass;state why. 3

(d) Using the necessary theorem explains how band structure is formed of a particular solid-state material. 3

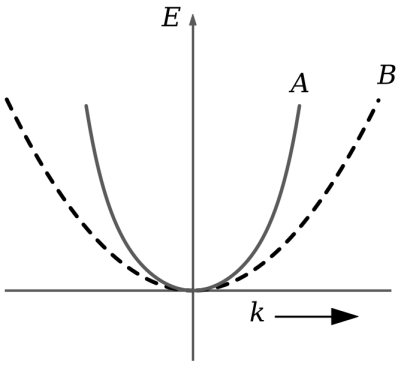


Figure 1: Two possible conduction bands A and B in the E versus k diagram (redrawn from the paper)

Term Test 2 (its questions 1 to 3 are 8(a) to 8(c))

[Course Code: EEE-421, Course Title: Solid State Devices, Credit: 3.0, Session: 2018-19, Total Marks: 10, Total Time: 30 minutes]

8. (a) A silicon pn junction at T = 300 K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{20}$ [*sic*: should be 5×10^{15}] cm^{-3}. Determine x_n, x_p and W. 3

(b) Determine the position of the Fermi level with respect to the valence band energy in p type GaAs at T = 300 K. The doping concentrations are [unclear: N_a] $= 5 \times 10^{19}$ [*sic*: should be 5×10^{16}] cm^{-3} and $N_d = 10^{15}$ [*sic*: should be 4×10^{15}] cm^{-3}. 4

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(c) Silicon at T = 300 K is doped with impurity concentrations of Nd = 5 × 1016 [*sic*: should be 10^{16}] /cm3 and Na = 2 × 1016 [*sic*: should be 10^{16}] /cm3. (a) What are the electron and hole mobilities? (b) Determine the conductivity and resistivity of the [unclear: material]. 3

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